

Olympiad Problems

Mathematics Initiatives in Nepal

August 16, 2026

1 Questions

1. Triangle MIN is isosceles with $MI = MN$. Medians $\overline{MB}$ and $\overline{CA}$ are perpendicular to each other, and $MB = CA = 12$. Find, with proof, the area of $\triangle MIN$.
2. You are given that x and y are real numbers.
 - (a) Given that $x + y$ and $x + y^2$ are rational numbers, is it always true that x and y are also rational numbers?
 - (b) Given that $x + y$, $x + y^2$, and $x + y^3$ are rational numbers, is it always true that x and y are also rational numbers?
3. For every positive integer n , express, in terms of n , the number of n -digit positive integers satisfying both of the following conditions:
 - No two consecutive digits are equal.
 - The last digit is prime.
4. Ram and Hari have created a program that, given a list of n numbers $a_1, a_2, \dots, a_n$, calculates a list $b_1, b_2, \dots, b_n$ such that, for each $1 \leq k \leq n$, the number b_k is the number of times a_k appears in the list. For example, the list

1, 2, 3, 1

outputs

2, 1, 1, 2.

Next, Ram asks Hari to run the program on a list of 2083 numbers. He then asks him to apply the program to the resulting list, and so on, until a number greater than or equal to k appears in the list. Find, with proof, the largest value of k for which, whatever the initial list of 2083 positive integers is, it is always possible for Hari to do what Ram asked.